DSA

Practical 7

Double Ended Queue

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Batch: A2

CODE:-

```
#include<iostream>
using namespace std;

#define SIZE 5
int dq_arr[SIZE];
int front=-1;
int rear=-1;

void enqueue_front(int val){
   if((front==0&&rear==SIZE-1)||front==rear+1){
      cout<<"Queue is Full."<<endl;
      return;
   }
   if(front==-1){
      front=rear=0;
   }else if(front==0){
      front=SIZE-1;
   }else{
      front--;
   }
   dq_arr[front]=val;
   cout<<val<<" is inserted."<<endl;
}

void enqueue_rear(int val){
   if((front==0&&rear==SIZE-1)||front==rear+1){
      cout<<"Queue is Full."<<endl;
      return;
   }
   if(front==-1){
```

```
        front=rear=0;
    }else if(rear==SIZE-1){
        rear=0;
    }else{
        rear++;
    }
    dq_arr[rear]=val;
    cout<<val<<" is inserted."<<endl;
}

void dequeue_front(){
    if(front==-1){
        cout<<"Queue is Empty."<<endl;
        return;
    }
    cout<<dq_arr[front]<<" is deleated."<<endl;
    if(front==rear){
        front=rear=-1;
    }else if(front==SIZE-1){
        front=0;
    }else{
        front++;
    }
}

void dequeue_rear(){
    if(front==-1){
        cout<<"Queue is Empty."<<endl;
        return;
    }
    cout<<dq_arr[rear]<<" is deleated."<<endl;
```

```
    if(front==rear){
        front=rear=-1;
    }else if(rear==0){
        rear=SIZE-1;
    }else{
        rear--;
    }
}

void display(){
    if(front==-1){
        cout<<"Queue is Empty."<<endl;
        return;
    }
    for(int i=front;;i=(i+1)%SIZE){
        cout<<dq_arr[i]<<" ";
        if(i==rear) break;
    }
    cout<<endl;
}

int main(){
    int choice;
    do {
        cout<<"\n--- Menu ---\n";
        cout<<"1. Enqueue Front\n";
        cout<<"2. Enqueue Rear\n";
        cout<<"3. Dequeue Front\n";
        cout<<"4. Dequeue Rear\n";
        cout<<"5. Display\n";
        cout<<"6. Exit\n";
```

```
cout<<"Enter choice: ";

if(!(cin >> choice)){
   cin.clear();
   cin.ignore(10000,'\n');
   cout<<"Invalid input. Try again."<<endl;
   continue;
}

switch(choice){
   case 1: {
      int val;
      cout<<"Enter value to insert at front: ";
      if(!(cin>>val)){
         cin.clear();
         cin.ignore(10000,'\n');
         cout<<"Invalid input."<<endl;
      } else {
         enqueue_front(val);
      }
      break;
   }
   case 2: {
      int val;
      cout<<"Enter value to insert at rear: ";
      if(!(cin>>val)){
         cin.clear();
         cin.ignore(10000,'\n');
         cout<<"Invalid input."<<endl;
      } else {
         enqueue_rear(val);
```

```
            }
            break;
        }
        case 3:
            dequeue_front();
            break;
        case 4:
            dequeue_rear();
            break;
        case 5:
            display();
            break;
        case 6:
            cout<<"Exiting."<<endl;
            break;
        default:
            cout<<"Invalid choice."<<endl;
      }
   } while(choice!=6);

   return 0;
}
```

OUTPUT:-

/*

--- Menu ---

1. Enqueue Front

2. Enqueue Rear

3. Dequeue Front

4. Dequeue Rear

5. Display

6. Exit

Enter choice: 1

Enter value to insert at front: 45

45 is inserted.

--- Menu ---

1. Enqueue Front

2. Enqueue Rear

3. Dequeue Front

4. Dequeue Rear

5. Display

6. Exit

Enter choice: 2

Enter value to insert at rear: 65

65 is inserted.

--- Menu ---

1. Enqueue Front

2. Enqueue Rear

3. Dequeue Front

4. Dequeue Rear

5. Display

6. Exit

Enter choice: 1

Enter value to insert at front: 85

85 is inserted.

--- Menu ---

1. Enqueue Front

2. Enqueue Rear

3. Dequeue Front

4. Dequeue Rear

5. Display

6. Exit

Enter choice: 2

Enter value to insert at rear: 95

95 is inserted.

--- Menu ---

1. Enqueue Front

2. Enqueue Rear

3. Dequeue Front

4. Dequeue Rear

5. Display

6. Exit

Enter choice: 5

85 45 65 95

--- Menu ---

1. Enqueue Front

2. Enqueue Rear

3. Dequeue Front

4. Dequeue Rear

5. Display

6. Exit

Enter choice: 4

95 is deleated.

--- Menu ---

1. Enqueue Front

2. Enqueue Rear

3. Dequeue Front

4. Dequeue Rear

5. Display

6. Exit

Enter choice: 3

85 is deleated.

--- Menu ---

1. Enqueue Front

2. Enqueue Rear

3. Dequeue Front

4. Dequeue Rear

5. Display

6. Exit

Enter choice: 5

45 65

--- Menu ---

1. Enqueue Front

2. Enqueue Rear

3. Dequeue Front

4. Dequeue Rear

5. Display

6. Exit

Enter choice: 6

Exiting.

*/